Worksheet: Effectiveness of liquid-liquid extraction

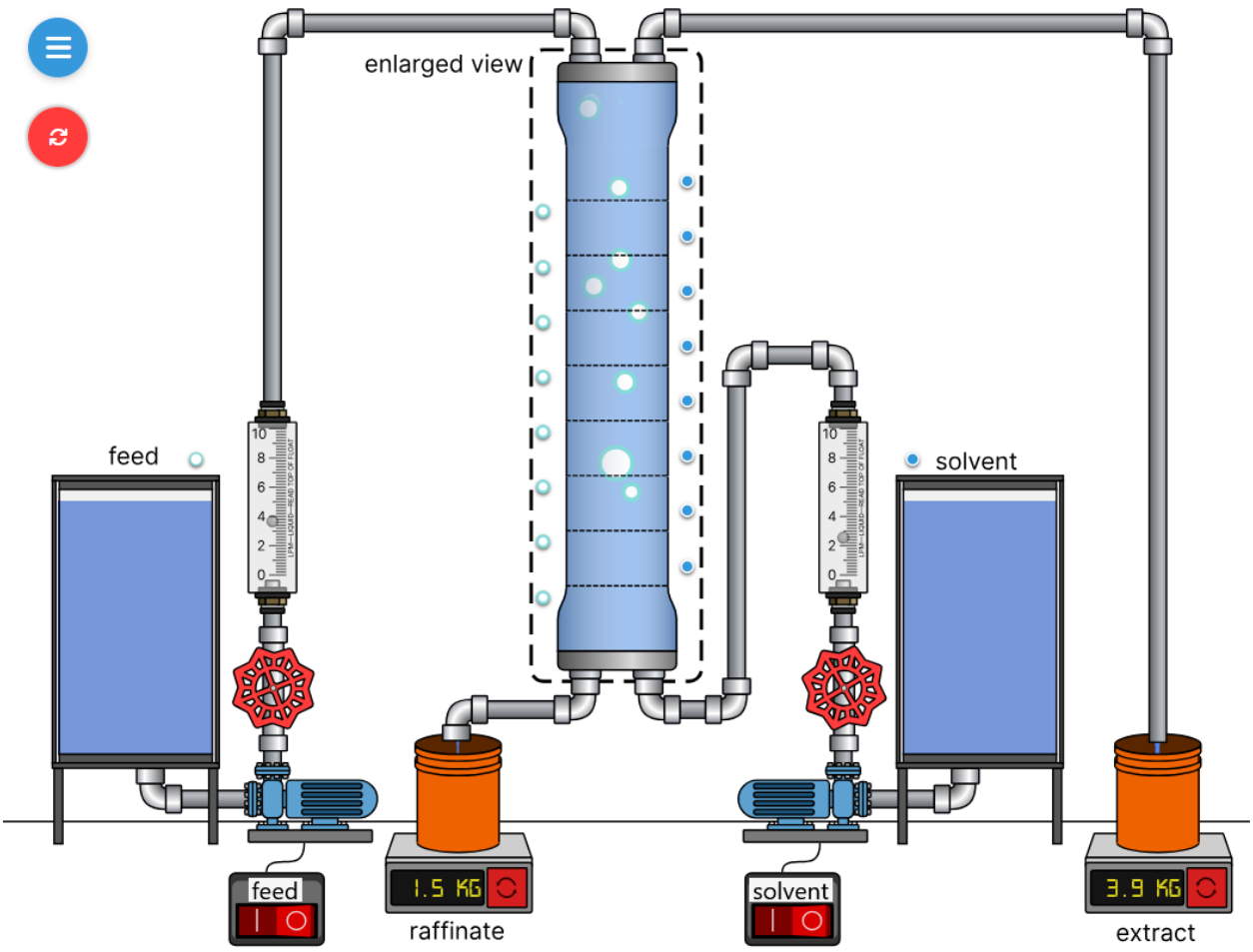
Name(s) _____

This experiment uses a liquid-liquid extraction process in which water as a solvent contacts an acetic acid/chloroform mixture in a steady-state counter flow experiment. Because acetic acid is much more soluble in water, it is extracted from the chloroform organic phase into the aqueous phase.

Student learning objectives

1. Be able to apply mass balances on a three-component system.
2. Determine the effects of the number of stages and stage efficiency on the separation process.
3. Be able to explain how changing the relative flow rates of feed and solvent affects the separation.
4. Be able to generate a Hunter-Nash analysis on a ternary phase diagram.

Equipment



The number of stages can be varied between one and eight, and the stage efficiency can be varied from 50% to 100%. In each stage, the aqueous solvent phase contacts the organic phase

(acetic acid, chloroform, and a small concentration of water) in counter-current flow. The phases leaving each stage are in phase equilibrium. The water extracts acetic acid solute from the organic feed. Thus, the extract contains the water solvent with acetic acid and chloroform. Chloroform is enriched in the raffinate because acetic acid and water have been removed from it.

S = solvent (blue) F= feed (white) E = Extract R = Raffinate

Note that the feed and solvent tanks are large relative to the size of the column (thus the column is enlarged in the simulation view). Thus, the tanks volumes do not change significantly during an experiment.

Questions to answer before starting the experiment

1. How will increasing the number of stages affect the separation?

2. How will decreasing stage efficiency affect the separation?

3. If the solvent flow rate decreases, how is the separation affected?

Experiment procedure:

Three distinct experiments are described below.

1. The mass percentages of the three components in the feed and their densities are:

   Table 1: Feed composition and density

Component	Mass percent	Density (kg/L)
Acetic acid	52.0	1.49
Chloroform	43.0	1.04
Water	5.0	1.00
Feed		

   The solvent only contains water.

   Assume the density of a mixture is $\rho_{mix} = 1/\sum(x_i/\rho_i)$, where x_i is the mass fraction and ρ_i is the density of component *i*. Calculate fluid density in the feed tank and record it in Table 1.

2. Select the edit internals button just above the feed tank to choose the number of stages and the stage efficiency. For each of the three experiments below, these values will be different. To change values after the start of an experiment, select the reset button, which resets the entire system.

Experiment #1: Effect of number of stages

1. Select the number of stages as 2 and the stage efficiency as 100%. Turn on the solvent pump by clicking the |/O switch and turn the solvent valve counterclockwise to start solvent flow into the extraction system. The column will fill with solvent. Once the column fills with solvent, turn on the feed pump by clicking the |/O switch (only possible after the column fills with solvent), and turn the feed valve counterclockwise to increase the feed flow rate. Select values where the feed flow rate is significantly higher than the solvent flow rate.
2. Record the volumetric flow rates for the feed and solvent and calculate the mass flow rates and record those values in Table 2.
3. Allow time for the system to reach steady state and liquid to accumulate in the extract and raffinate tanks. Then determine the mass flow rates of the two exit streams by recording the mass change in the buckets while recording the time elapsed with your phone. Record values in Table 2.

Table 2: Two stages, 100% efficient

Tank	Volumetric flow rate (L/min)	Density (kg/L)	Elapsed time (s)	Mass change (kg)	Mass flow rate (kg/min)
Feed					
Solvent		1.00			
Extract					
Raffinate					

4. Do an overall mass balance on the system.

Total mass flow rate in _____

Total mass flow rate out _____

Difference (In – Out) _____

5. The dots on the sides of the column measure the compositions leaving each stage. On the left are the organic phase compositions and on the right are the aqueous phase compositions. Measure the composition of the raffinate stream by moving your mouse over the bottom dot on the left side of the column. Measure the composition of the extract stream by moving your mouse over the top dot on the right side of the column. The top stage in the column is stage 1. Record these compositions in Table 3

Table 3 Mass fractions for two stages, 100% efficient

	Mass fractions feed side			Mass fractions solvent side		
Stage	Acetic acid	Chloroform	Water	Acetic acid	Chloroform	Water
1						
2						

6. Repeat step 1 to 4 but select the number of stages as 8 and the stage efficiency as 100%. Set the same volumetric flow rates for the feed and solvent. Record the feed flow rates in Table 4. Record the mass fractions in Table 5.

Table 4: Eight stages, 100% efficient

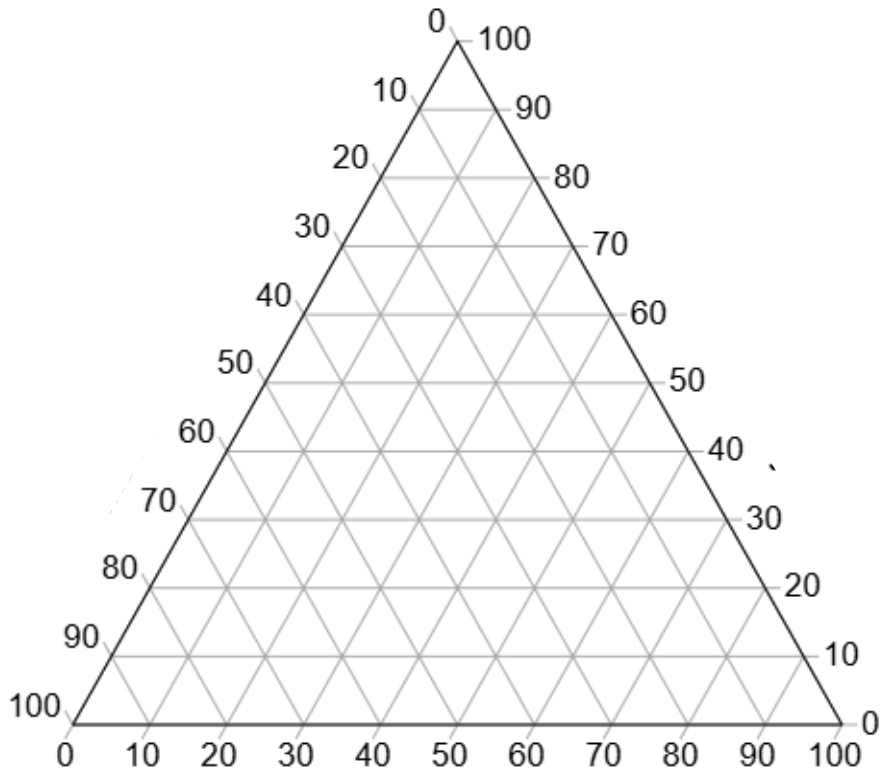
Tank	Volumetric flow rate (L/min)	Density (kg/L)	Elapsed time (s)	Mass change (kg)	Mass flow rate (kg/min)
Feed					
Solvent		1.00			
Extract					
Raffinate					

Table 5 Mass fractions for eight stages, 100% efficient

	Mass fractions feed side			Mass fractions solvent side		
Stage	Acetic acid	Chloroform	Water	Acetic acid	Chloroform	Water
1						
2						
3						
4						
5						
6						
7						
8						

7. Compare the results for two stages and eight stages. What is different? Why?

8. Use the results for 8 stages to make a Hunter-Nash plot on a ternary phase diagram. Use the data in Table 5 to draw tie lines and the phase envelope. The streams leaving each stage (e.g., E_2 and R_2) are in equilibrium. Tie lines connect two composition points that are in equilibrium on a ternary phase diagram. Points connected by tie lines are on the phase envelope of a ternary phase diagram. Locate the six points (exit stream concentrations) on the ternary phase diagram below and draw as much of the phase envelope as you can.



Experiment #2: Effect of stage efficiency

1. Repeat the measurements as in Experiment #1 for eight stages with a stage efficiency of 60%. Record the results in Tables 6 and 7.

Table 6: Eight stages, 60% efficient

Tank	Volumetric flow rate (L/min)	Density (kg/L)	Elapsed time (s)	Mass change (kg)	Mass flow rate (kg/min)
Feed					
Solvent		1.00			
Extract					
Raffinate					

Table 7 Mass fractions for eight stages, 60% efficient

	Mass fractions feed side			Mass fractions solvent side		
Stage	Acetic acid	Chloroform	Water	Acetic acid	Chloroform	Water
1						
2						
3						
4						
5						
6						
7						
8						

2. Compare the results in Tables 5 and 7. What can you conclude?

Experiment #3: Effect of relative flow rates of feed and solvent

1. Select the number of stages as 5 and the stage efficiency as 100%. Turn on the solvent pump by clicking the 1/0 switch and turn the solvent valve counterclockwise to start solvent flow into the extraction system. Select a flow rate in the middle of the flowmeter range. The column will fill with solvent. Once the column fills with solvent, turn on the feed pump (only possible after the column fills with solvent), and turn the feed valve counterclockwise to increase the feed flow rate.
2. Select a feed flow rate at the low end of the flow meter.
3. Record the volumetric flow rates for the feed and solvent and calculate the mass flow rates and record those values in Table 8.
4. Allow time for the system to reach steady state and liquid to accumulate in the extract and raffinate tanks. Then determine the mass flow rates of the two exit streams by recording the mass change in the buckets while recording the time elapsed with your phone. Record values in Table 8.

Table 8: Five stages, 100% efficient, low feed flow rate

Tank	Volumetric flow rate (L/min)	Density (kg/L)	Elapsed time (s)	Mass change (kg)	Mass flow rate (kg/min)
Feed					
Solvent		1.00			
Extract					
Raffinate					

5. Do an overall mass balance on the system.

Total mass flow rate in _____

Total mass flow rate out _____

Difference (In – Out) _____

6. The dots on the sides of the column measure the compositions leaving each stage. On the left are the organic phase compositions and on the right are the aqueous phase compositions. Measure the composition of the raffinate stream by moving your mouse over the bottom dot on the left side of the column. Measure the composition of the extract stream by moving your mouse over the top dot on the right side of the column. The top stage in the column is stage 1. Record these compositions in Table 9.

Table 9 Mass fractions for five stages, 100% efficient, low feed flow rate

	Mass fractions feed side			Mass fractions solvent side		
Stage	Acetic acid	Chloroform	Water	Acetic acid	Chloroform	Water
1						
2						
3						
4						
5						

7. Repeat these measurements by changing the feed flow rate to the middle of the flowmeter range and record the results in Tables 10 and 11.

   Table 10: Five stages, 100% efficient, medium feed flow rate

Tank	Volumetric flow rate (L/min)	Density (kg/L)	Elapsed time (s)	Mass change (kg)	Mass flow rate (kg/min)
Feed					
Solvent		1.00			
Extract					
Raffinate					

8. Do an overall mass balance on the system.

   Total mass flow rate in _____

   Total mass flow rate out _____

   Difference (In – Out) _____

Table 11 Mass fractions for five stages, 100% efficient, medium feed flow rate

	Mass fractions feed side			Mass fractions solvent side		
Stage	Acetic acid	Chloroform	Water	Acetic acid	Chloroform	Water
1						
2						
3						
4						
5						

9. Repeat these measurements by changing the feed flow rate to near the top of the flowmeter range and record the results in Tables 12 and 13.

Table 12: Five stages, 100% efficient, high feed flow rate

Tank	Volumetric flow rate (L/min)	Density (kg/L)	Elapsed time (s)	Mass change (kg)	Mass flow rate (kg/min)
Feed					
Solvent		1.00			
Extract					
Raffinate					

10. Do an overall mass balance on the system.

Total mass flow rate in _____

Total mass flow rate out _____

Difference (In – Out) _____

Table 13 Mass fractions for five stages, 100% efficient, high feed flow rate

	Mass fractions feed side			Mass fractions solvent side		
Stage	Acetic acid	Chloroform	Water	Acetic acid	Chloroform	Water
1						
2						
3						
4						
5						

11. Discuss the effects of the relative flow rates of feed and solvent on the separation and on the flow rates of raffinate and extract.

Questions to answer

1. Where might these measurements have errors?

2. What safety precautions would you take to conduct this experiment in the laboratory?

3. At what conditions (number of stages, flow rates) would you recommend running this liquid-liquid extraction? Why?